

Dr. FILOTHEOU, Alexandros

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Robotics Engineer with eight years designing and deploying applications including autonomous navigation, state estimation, and multi-robot orchestration on ground and aerial platforms in real-world environments. Proven track record leading complex multi-partner integrations from architecture to implementation and field deployment across countries. Backed by IEEE publications and single-authored papers in IROS'24 and R&AS.

Skills

English	Native / Fluent (IELTS 8.5 - C2 Proficiency)
Languages	C/C++, Python, shell, MATLAB/Octave
Robotics/OS	Linux, ROS/ROS 2
Tools/Frameworks	git, Docker, Behavior Trees, Gazebo, Eigen, CI/CD, Qt/Tkinter
Sensors/Hardware	Ouster 3D/Hokuyo 2D LiDARs, SICK Visionary-T, Astra, Realsense, ZED2, Impinj R420

Highlights

- Deployed €10M robotics construction project [RoBétArmé](#) after 2 years of continuous integration between 12 partners at [EDF's Hermillon hydroelectric powerplant](#) in cooperation with Bouygues Construction
- Designed and deployed full-stack [autonomous social robot](#) (NLP, Navigation, SLAM, RFID, HRI+GUI) at [AMTh](#) museum, engaging 2,000+ visitors annually since 2023
- Identified bottlenecks → reduced deployment time >10x across fleet of additive manufacturing robots
- Engineered 50% boost in localisation accuracy for fleet of RFID-inventorying robots → achieved cm accuracy
- Resolved cross-machine ROS/ROS 2 interoperability for entire 12-partner consortium → unblocked all teams
- One of only eighteen authors of single-authored publications presented at [IEEE IROS'24](#) (1,587 total)

Representative work

Demos	State Estimation	Lidar Odometry	Publications
Integration [1] · [2] · [3] SLAM/Navigation Museum Robot Control is Fun	Video Paper Code	Video Paper Code	Estimation [Local] · [Global] Localisation/Control (solo) Multi-robot MPC Navigation Survey

Professional Experience

- Senior Robotics Software Engineer** · [Intermodalics](#), Leuven BE Feb 2026 – Present
- At [KION Group AG](#): Bridging VDA5050 fleet manager with ROS 2 mission behaviours within global team developing mass-production industrial autonomous forklift AMRs for warehouses
- Integration and Deployment Engineer** · [CERTH/ITI](#), Thessaloniki GR Sep 2023 – Jan 2026
- Architected and implemented software integration for 12-partner EU Horizon project [RoBétArmé](#): Docker infrastructure, CI/CD pipelines, and cross-machine ROS/ROS 2 interoperability via [Zenoh](#) & [ros1_bridge](#)
 - Led Task Planning & Behaviour Tree Orchestration for concrete- and [metal-additive](#) manufacturing robots
 - Defined development, integration, and deployment principles for all 12 SW teams and >50 Dockerised ROS/ROS 2 packages; ensured code quality via googletest and cplint
- Robotics Research Engineer** · ECE Dept., Aristotle University of Thessaloniki GR Sep 2018 – Mar 2023
- Sole developer of all robotics SW for R&D projects [RELIEF](#) & [CultureId](#): autonomous ground (Turtlebot 2, Robotnik RB1) and aerial platforms for RFID inventorying; social robot for interactive museum engagement
 - Delivered & deployed 2D/3D SLAM and autonomous navigation for warehouses, libraries, and museums
 - Developed and integrated codebases producing 18+ publications in top-tier IEEE journals & conferences
- Computer Vision Engineer** · [PANDORA Robotics](#), Thessaloniki GR Oct 2013 – Jul 2014
- Gained 2nd place in 2015 RoboCup Rescue competition developing Kinect-based [C++ wall-hole detection](#)

Open Source Contributions to ROS 2

- [hitch_estimation_apriltag_array](#) · Estimation of angle between vehicle and hitched trailer
- [pointcloud_to_ply](#) · Transformation of pointcloud into mesh and storage in .ply or .obj format

Educational Background

- Doctorate** · Electrical & Computer Engineering · Aristotle University of Thessaloniki Sep 2018 – Jun 2023
- Master of Science** · Systems, Control, and Robotics · KTH Royal Inst. of Technology Sep 2015 – Jun 2017
Thesis: Decentralised MPC for multi-robot constrained navigation [ECC'18, IJC'18]
- Diploma** · Electrical & Computer Engineering · Aristotle University of Thessaloniki Sep 2005 – Jul 2013